Prelim 2025: Answer Key

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Question 5 (*25 points*)

a) State the consumer's problem. Characterize the solution with first-order conditions. Let λ_t be the Lagrange multiplier associated to the budget constraint, and μ_t the Lagrange multiplier associated to the borrowing constraint. Define a competitive equilibrium for this economy. **Hint:** Remember that the financial markets are international, so you do not need to include a market clearing condition for bonds.

Solution:

The consumer's problem is

$$\max_{\{C_t, K_{t+1}, L_{t+1}, B_{t+1}\}_{t\geq 0}} \sum_{t=0}^{\infty} \beta^t \ln C_t, \qquad \beta \equiv \frac{1}{1+\rho} = \frac{1}{1+r},$$

subject to, for every $t \geq 0$,

$$C_t + K_{t+1} + Q_t L_{t+1} + (1+r)B_t \leq F(K_t, L_t) + B_{t+1} + Q_t L_t$$
$$B_{t+1} \leq \bar{B}$$

and $K_0, B_0 > 0$, $L_0 = 1$ given.

Form the Lagrangian

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t \Big\{ \ln C_t + \lambda_t \big[F(K_t, L_t) + B_{t+1} + Q_t L_t - C_t - K_{t+1} - Q_t L_{t+1} - (1+r)B_t\big] + \mu_{t+1}(\bar{B} - B_{t+1}) \Big\}.$$

The first-order conditions are

$$(C_t): \quad \frac{1}{C_t} = \lambda_t \tag{1}$$
$$(K_{t+1}): \quad \lambda_t = \beta \lambda_{t+1} F_K(K_{t+1}, L_{t+1}) \tag{2}$$
$$(L_{t+1}): \quad \lambda_t Q_t = \beta \lambda_{t+1} \big[F_L(K_{t+1}, L_{t+1}) + Q_{t+1}\big] \tag{3}$$
$$(B_{t+1}): \quad \lambda_t = \beta(1+r)\lambda_{t+1} + \beta\mu_{t+1} = \lambda_{t+1} + \beta\mu_{t+1} \tag{4}$$

together with the complementary slackness condition $\mu_t(\bar{B} - B_t) = 0$, $\mu_t \geq 0$, and a transversality condition. We used $\beta(1+r) = 1$ in (4).

A *competitive equilibrium* is a sequence of allocations $\{C_t, K_{t+1}, L_{t+1}, B_{t+1}\}_{t\geq 0}$ and a land price sequence $\{Q_t\}_{t\geq 0}$ such that, given prices, the allocation solves the consumer's problem above, and the land market clears every period:

$$L_t = 1 \quad \text{for all } t \geq 0.$$

The bond market is international and does not need to clear.

b) A steady state equilibrium is an equilibrium in which $C_t = C_{ss}$, $B_t = B_{ss}$, $Y_t = Y_{ss}$, $Q_t = Q_{ss}$ and $K_t = K_{ss}$ every period. Using the definition of a competitive equilibrium and the FOC's of the consumer's optimization problem, characterize a steady state equilibrium, given a borrowing level $B_{ss} < \bar{B}$. Do K_{ss}, Y_{ss} and Q_{ss} depend on the level of B_{ss}?

Solution:

Since $B_{ss} < \bar{B}$, the borrowing constraint is slack and $\mu_t = 0$ in steady state. Equation (4) then gives $\lambda_t = \lambda_{t+1}$, consistent with constant consumption via (1). Equations (2) and (3) reduce to

$$F_K(K_{ss}, 1) = \frac{1}{\beta} = 1 + r \tag{5}$$

$$Q_{ss} = \beta\big[F_L(K_{ss}, 1) + Q_{ss}\big] \implies Q_{ss} = \frac{F_L(K_{ss}, 1)}{r} \tag{6}$$

Evaluating the budget constraint in steady state with $L_t = 1$, $K_{t+1} = K_t = K_{ss}$ and $B_{t+1} = B_t = B_{ss}$,

$$C_{ss} = F(K_{ss}, 1) - K_{ss} - rB_{ss}. \tag{7}$$

Equation (5) pins down K_{ss} purely from the technology and the world interest rate. Hence $Y_{ss} = F(K_{ss}, 1)$ and (by (6)) Q_{ss} are independent of B_{ss}. Only C_{ss} depends on B_{ss}: a larger debt level reduces steady-state consumption one-for-one through the interest payment rB_{ss}.

c) Assume that $K_0 = K_{ss}$, $B_0 = B_{ss} < \bar{B}$, and $L_0 = 1$ (so the economy is at its steady state from period $t = 0$). Suppose that there is an unexpected positive shock on the debt limit; i.e., $\bar{B}' = \bar{B} + \Delta$, with $\Delta > 0$. Does this shock have any effect on consumption, output and land prices? What if $\bar{B}' = \bar{B} - \Delta$ with $\Delta \in (0, \bar{B} - B_0)$? Would it change your results if there was no borrowing constraint?

Solution:

In all three sub-cases the borrowing constraint is slack at the original equilibrium ($B_0 = B_{ss} < \bar{B}$). The unconstrained FOCs are satisfied at the original allocation $\{C_{ss}, K_{ss}, B_{ss}, Q_{ss}\}$, and that allocation remains feasible under the new constraint.

(i) Looser limit $\bar{B}' = \bar{B} + \Delta$: the new constraint is even slacker, so the steady-state allocation continues to satisfy all optimality and market-clearing conditions. **No effect** on C_t, Y_t, Q_t. The consumer was already free to borrow more and chose not to: with $\rho = r$ she desires perfect consumption smoothing and has no demand for additional borrowing at the existing C_{ss}.

(ii) Tighter limit $\bar{B}' = \bar{B} - \Delta$ with $\Delta \in (0, \bar{B} - B_0)$: since $\bar{B}' > B_0 = B_{ss}$, the new constraint still does not bind at the original allocation. **No effect.**

(iii) No borrowing constraint at all: same conclusion. Without the constraint, the Euler equation together with $\beta(1 + r) = 1$ delivers the same steady-state allocation.

d) Suppose that $\bar{B}' = \bar{B} - \Delta$, and that $\Delta > \bar{B} - B_0$ (so that the initial borrowing level exceeds the borrowing limit $\bar{B}$). What happens with equilibrium output, Y_t, and the equilibrium price of land, Q_t? **Hint:** Guess that there exists an equilibrium in which the borrowing constraint binds every period, i.e. $\mu_t > 0$ $\forall t$. First, show that in such an equilibrium, $\{C_t\}_{t=0}^{\infty}$ increases over time. Then, show that $K_{t+1} < K_{ss}$ for all t. Finally, iterate forward the optimality condition for land to obtain the price of land as a function of the future path of consumption and capital.

Solution:

Guess that $\mu_t > 0$ for every $t \geq 1$, so that the constraint binds and $B_t = \bar{B}'$ for $t \geq 1$.

Step 1: consumption growth. Combining (1) with (4),

$$\frac{1}{C_t} = \frac{1}{C_{t+1}} + \beta\mu_{t+1}.$$

The multiplier μ_{t+1} is determined by this equation given the consumption path. If $\mu_{t+1} > 0$ (as we verify in Step 4 below), then $C_{t+1} > C_t$. Intuitively, the binding constraint forces deleveraging today: with $\rho = r$ the consumer would like to spread the resource loss across all dates by borrowing, but the constraint prevents this, so marginal utility is high in early periods and falls over time.

Step 2: $K_{t+1} < K_{ss}$. From (2) and (4),

$$F_K(K_{t+1}, 1) = \frac{\lambda_t}{\beta\lambda_{t+1}} = \frac{\lambda_{t+1} + \beta\mu_{t+1}}{\beta\lambda_{t+1}} = (1 + r) + \frac{\mu_{t+1}}{\lambda_{t+1}} > 1 + r = F_K(K_{ss}, 1).$$

Since F is concave, F_K is decreasing in K, so $K_{t+1} < K_{ss}$. The high shadow value of funds makes the consumer reluctant to allocate resources to investment: capital must earn a marginal return above $1 + r$ to compete with the shadow return on relaxing the borrowing constraint. It follows that $Y_t = F(K_t, 1) < Y_{ss}$ for all $t \geq 1$.

Step 3: Land price. Define the (constraint-adjusted) discount factor $\Lambda_{t,t+s} \equiv \beta^s \lambda_{t+s}/\lambda_t$. Iterating (3) forward and imposing the no-bubble condition $\lim_{s\to\infty} \Lambda_{t,t+s} Q_{t+s} = 0$,

$$Q_t = \sum_{s=1}^{\infty} \Lambda_{t,t+s} F_L(K_{t+s}, 1). \tag{8}$$

From (4), $\beta\lambda_{t+1}/\lambda_t = 1/(1 + \beta\mu_{t+1}/\lambda_{t+1}) < 1$ whenever $\mu_{t+1} > 0$, so $\Lambda_{t,t+s} < \beta^s$ for all $s \geq 1$ along the binding path. Combined with $F_L(K_{t+s}, 1) < F_L(K_{ss}, 1)$ (assuming the standard $F_{KL} > 0$, as in Cobb-Douglas), each term in (8) is below its steady-state counterpart, so

$$Q_t < \sum_{s=1}^{\infty} \beta^s F_L(K_{ss}, 1) = Q_{ss}.$$

Step 4: verifying the guess. Combining (2) and (4),

$$\mu_{t+1} = \lambda_{t+1}\big[F_K(K_{t+1}, 1) - (1 + r)\big],$$

so by concavity of F and $F_K(K_{ss}, 1) = 1 + r$,

$$\mu_{t+1} > 0 \iff K_{t+1} < K_{ss}, \quad \mu_{t+1} = 0 \iff K_{t+1} = K_{ss}, \quad \mu_{t+1} < 0 \iff K_{t+1} > K_{ss}.$$

Since $\mu \geq 0$, the third case is ruled out: any equilibrium must have $K_t \leq K_{ss}$ at every $t \geq 1$.

Suppose, contrary to the guess, $\mu_{t^*} = 0$ for some $t^* \geq 1$, with t^* the first such date. From the equivalence above, $K_{t^*} = K_{ss}$. From (4), $C_{t^*-1} = C_{t^*}$. Use the budget at $t^* - 1$ to compute C_{t^*-1}:

Case $t^* \geq 2$. The budget reads $C_{t^*-1} + K_{t^*} = F(K_{t^*-1}, 1) - r\bar{B}'$. Since $\mu_{t^*-1} > 0$, $K_{t^*-1} < K_{ss}$, so $F(K_{t^*-1}, 1) < F(K_{ss}, 1)$. With $K_{t^*} = K_{ss}$,

$$C_{t^*-1} = F(K_{t^*-1}, 1) - r\bar{B}' - K_{ss} < F(K_{ss}, 1) - r\bar{B}' - K_{ss} = C^*,$$

where $C^* \equiv F(K_{ss}, 1) - K_{ss} - r\bar{B}'$.

Case $t^* = 1$. The budget reads $C_0 + K_1 = F(K_{ss}, 1) + \bar{B}' - (1+r)B_0$. With $K_1 = K_{ss}$,

$$C_0 = F(K_{ss}, 1) - K_{ss} + \bar{B}' - (1+r)B_0 = C^* - (1+r)(B_0 - \bar{B}') < C^*,$$

using $B_0 > \bar{B}'$.

In either case, $C_{t^*} = C_{t^*-1} < C^*$. Now apply the budget at t^* (with $K_{t^*} = K_{ss}$ and $B_{t^*+1} = \bar{B}'$):

$$K_{t^*+1} = F(K_{ss}, 1) - r\bar{B}' - C_{t^*} = K_{ss} + (C^* - C_{t^*}) > K_{ss}.$$

By the equivalence above, $K_{t^*+1} > K_{ss}$ would force $\mu_{t^*+1} < 0$, contradicting non-negativity. Hence $\mu_{t^*} = 0$ is impossible, and $\mu_t > 0$ for all $t \geq 1$.

Summary. The tightening of $\bar{B}$ that forces deleveraging triggers a recession: C_t jumps down on impact and rises back toward its new (lower) long-run level, capital and output are depressed below the original steady-state values, and the price of land falls. The decline in Q_t reflects two reinforcing effects: a steeper effective discount (the binding constraint inflates the marginal value of current funds) and a lower marginal product of land along the transition (because $K_t < K_{ss}$).

e) Suppose now that instead of the borrowing constraint (1), consumers must now collateralize their loans with the value of their land holdings. Namely,

$$B_{t+1} \leq Q_t L_{t+1}.$$

Explain in words how this economy differs from the one in part d). In particular, explain whether the endogenous response of the price of land, Q_t, amplifies or dampens the initial shock. Is this an example of fire sales?

Solution:

In part d) the borrowing limit was an exogenous constant, so a fall in Q_t had no feedback on borrowing capacity. With a collateral constraint, the borrowing limit is the market value of the consumer's land holdings, which is itself determined endogenously.

Consider the same kind of shock that pushed the consumer against her debt limit. As in d), the binding constraint depresses K_{t+1} and the marginal utility of current consumption is high, so by (8), Q_t falls. But now a lower Q_t *directly tightens* the borrowing constraint: $B_{t+1} \leq Q_t L_{t+1}$ shrinks one-for-one with Q_t. The consumer must deleverage further, pushing K_{t+2} and future F_L down, which depresses Q_{t+1}, which further tightens future borrowing, and so on. The endogenous response of Q_t **amplifies** the initial shock. Under the exogenous limit of d) this feedback is absent.

Is this a fire sale? It captures the central amplification mechanism of fire-sale models in the spirit of Lorenzoni/Kiyotaki-Moore: an adverse shock triggers a fall in asset prices, the price decline tightens the collateral constraint of credit-dependent agents, and the resulting forced adjustment pushes prices down further, in a self-reinforcing loop. Strictly speaking, the canonical fire-sale story also involves reallocating the asset from a high-valuation user to a low-valuation user (the pecuniary externality runs through the marginal user). Here, with a representative agent, the asset is not reallocated across heterogeneous types, so the price falls through the direct effect of higher effective discounting and lower future F_L rather than through reallocation. Both readings are reasonable; the key economic point is the collateral feedback loop that distinguishes part e) from part d).

Question 6 (*25 points*)

a) Argue that if the consumers' endowment is sufficiently large, the gross rate of return on the risk-free bond is equal to 1 (i.e., the net return, r, is zero).

Solution:

Consumers have access to a storage technology with gross return 1, so the gross return on the risk-free bond cannot be below 1. They are also endowed with a "large" amount of the consumption good in $t = 1$, meaning more endowment than the entrepreneurs would want to borrow even at the lowest possible interest rate. Therefore some consumers strictly prefer to save in storage. Competition then implies that the equilibrium gross interest rate cannot exceed 1 either. Hence $r = 0$.

b) Suppose that entrepreneurs face no borrowing constraints. State the optimization problems of an entrepreneur and a consumer. Show that the equilibrium capital price is $q_1 = A$ and the entrepreneurs buy k^*, where $G'(\bar{k} - k^*) = A$.

Solution:

The entrepreneur's problem is

$$\begin{aligned} \max_{c_2, k \geq 0, b_2} \quad & E[c_2] \\ \text{s.t. } & q_1 k \leq N_1 + b_2 \\ & c_2 + b_2 \leq Ak \end{aligned}$$

Monotonicity implies that both constraints bind. From the budget constraint, $b_2 = q_1 k - N_1$ (and one verifies $b_2 \leq Ak$ ex post). Substituting into the second constraint,

$$c_2 = (A - q_1)k + N_1,$$

so the problem reduces to

$$\max_{k \geq 0} (A - q_1)k + N_1.$$

The FOC is $A - q_1 \leq 0$, giving the demand

$$k = \begin{cases} 0 & \text{if } q_1 > A \\ x \in [0, \infty) & \text{if } q_1 = A \\ \infty & \text{if } q_1 < A. \end{cases}$$

The consumer solves

$$\begin{aligned} \max_{\tilde{c}_2, \tilde{k}, \tilde{s} \geq 0, b_2} \quad & E[\tilde{c}_2] \\ \text{s.t. } & b_2 + \tilde{s} \leq e_1 + q_1(\bar{k} - \tilde{k}) \\ & \tilde{c}_2 \leq e_2 + G(\tilde{k}) + b_2 + \tilde{s} \end{aligned}$$

where $\tilde{s}$ denotes storage. Monotonicity and $r = 0$ simplify the problem to

$$\max_{\tilde{k} \geq 0} e_2 + G(\tilde{k}) + e_1 + q_1(\bar{k} - \tilde{k}), \qquad \text{FOC: } q_1 = G'(\tilde{k}).$$

The equilibrium price cannot satisfy $q_1 < A$ (entrepreneur demand would be infinite). Suppose $q_1 > A$: then $k = 0$ and $\tilde{k} = \bar{k}$, but $G'(\bar{k}) = 0 < A$ contradicts the consumer's FOC. Therefore $q_1 = A$, $G'(\bar{k} - k^*) = A$, and $k = k^*$.

c) Suppose that the entrepreneurs cannot borrow at all, so $q_1 k_2 \leq N_1$. Find the equilibrium price and allocation, show that $q_1 < A$ in equilibrium and that the expected utility of the entrepreneur is $\frac{A}{q_1} N_1$. Show that q_1 is increasing in N_1.

Solution:

The entrepreneur now solves

$$\begin{aligned} \max_{c_2, k \geq 0} \; & E[c_2] \\ \text{s.t. } & q_1 k \leq N_1 \\ & c_2 \leq Ak. \end{aligned}$$

Trivially $k = N_1/q_1$. The consumer's problem is unchanged, so the equilibrium is

$$k = \frac{N_1}{q_1}, \qquad q_1 = G'\left(\bar{k} - \frac{N_1}{q_1}\right).$$

Since $G''(\cdot) < 0$, the RHS is strictly decreasing in q_1, so the equilibrium is unique. Moreover, $N_1 < Ak^*$ implies $k = N_1/q_1 < k^*$, which by the consumer's FOC implies $q_1 = G'(\bar{k} - k) < G'(\bar{k} - k^*) = A$.

When N_1 rises, the entrepreneurs' demand for capital N_1/q_1 rises, the consumers hold less capital, their marginal product G' rises, and so does the equilibrium price. Hence q_1 is strictly increasing in N_1 for $N_1 < Ak^*$.

The utility of the entrepreneur is

$$c_2 = Ak = A \cdot \frac{N_1}{q_1}.$$

d) Assume that the entrepreneurs' initial net worth N_0 is sufficiently large so that the constraint $(q_1(a) + a)k_1 - b_1 \geq 0$ is not binding, and that they face no borrowing limit in period 1. Show that asset prices are $q_0 = 2A$ and $q_1(a) = A$ for all a, and that the capital owned by the entrepreneurs is constant at k^*.

Solution:

The entrepreneur's problem becomes

$$\begin{aligned} \max_{c_2, k_1, k_2(a) \geq 0, b_1, b_2(a)} \; & E[c_2] \\ \text{s.t. } & q_0 k_1 \leq N_0 + b_1 \\ & q_1(a) k_2(a) + b_1 \leq (q_1(a) + a) k_1 + b_2(a) \\ & c_2(a) + b_2(a) \leq A k_2(a). \end{aligned}$$

Following the same simplifications as in b), the problem reduces to

$$\max_{k_1, k_2(a) \geq 0} E\big[(A - q_1(a)) k_2(a) + (q_1(a) + a - q_0) k_1 + N_0\big],$$

with FOCs

$$(k_1): \quad E[q_1(a) + a] \leq q_0, \qquad (k_2(a)): \quad A \leq q_1(a).$$

Positive and finite demand requires both with equality, so $q_1(a) = A$ for every a and $q_0 = E[q_1(a) + a] = 2A$.

The consumer's analogous problem reduces to

$$\max E\big[G(\tilde{k}_2(a)) + e_1 + G(\tilde{k}_1) + q_1(a)(\tilde{k}_1 - \tilde{k}_2(a)) + e_0 + q_0(\bar{k} - \tilde{k}_1)\big],$$

with FOCs $G'(\tilde{k}_1) + E[q_1(a)] = q_0$ and $G'(\tilde{k}_2(a)) = q_1(a)$. Substituting $q_1(a) = A$, $q_0 = 2A$ gives $k_1 = k_2 = k^*$ with $G'(\bar{k} - k^*) = A$.

e) Now suppose that the entrepreneurs cannot borrow at date 1 and that $N_1(a) < Ak^*$ for all a, but they are unconstrained at date 0. Argue informally that now the equilibrium asset price q_1 is weakly increasing in a. **Hint:** Check your answer to question c).

Solution:

From part c), holding a fixed, q_1 is increasing in net worth N_1. Net worth at date 1 is $N_1(a) = (q_1(a) + a)k_1 - b_1$, which is increasing in a (higher productivity raises both the dividend a and the resale price $q_1(a)$). Hence $q_1(a)$ inherits this monotonicity: $q_1(a)$ is weakly increasing in a.

f) Keep the assumption from question e). Write the entrepreneur's problem at date 0. Derive the first-order conditions for the entrepreneur's problem. Derive an expression for the equilibrium price q_0 as a function of $q_1(a)$. **Hint:** Remember that now $N_1(a) = (q_1(a) + a)k_1 - b_1$.

Solution:

The entrepreneur's problem at date 0 is

$$\begin{aligned} \max_{c_2, k_1, k_2(a) \geq 0, b_1} \quad & E[c_2] \\ \text{s.t. } & q_0 k_1 \leq N_0 + b_1 \\ & q_1(a)k_2(a) + b_1 \leq (q_1(a) + a)k_1 \\ & c_2(a) \leq Ak_2(a). \end{aligned}$$

From part c), once at date 1 the entrepreneur's continuation utility is $AN_1(a)/q_1(a)$. Using $N_1(a) = (q_1(a) + a)k_1 - b_1$ and $b_1 = q_0 k_1 - N_0$,

$$\max_{k_1} E\left[A \cdot \frac{(q_1(a) + a)k_1 - q_0 k_1 + N_0}{q_1(a)}\right].$$

The FOC in k_1 (interior solution) is

$$E\left[\frac{A}{q_1(a)}\left(q_1(a) + a - q_0\right)\right] = 0,$$

which rearranges to

$$q_0 = \frac{E\left[\frac{A}{q_1(a)}(q_1(a) + a)\right]}{E\left[\frac{A}{q_1(a)}\right]}. \tag{9}$$

g) Prove that

$$q_0 \leq E[a + q_1(a)]$$

with strict inequality if $q_1(a)$ is not constant. Why is the date 0 price of capital below its present discounted value, despite the fact that entrepreneurs have linear utility and are unconstrained? Why do entrepreneurs not borrow and leverage their investment more to profit from the positive expected return? **Hint:** Remember that if x and y are two random variables, then $cov(x, y) = E[xy] - E[x]E[y]$.

Solution:

Apply the hint to the numerator of (9):

$$E\left[\frac{A}{q_1(a)}(q_1(a)+a)\right] = \operatorname{cov}\left(\frac{A}{q_1(a)}, q_1(a)+a\right) + E\left[\frac{A}{q_1(a)}\right] E[q_1(a)+a].$$

Substituting into (9),

$$q_0 = E[a+q_1(a)] + \frac{\operatorname{cov}\left(\frac{A}{q_1(a)},\ q_1(a)+a\right)}{E\left[\frac{A}{q_1(a)}\right]}.$$

By part e), $q_1(a)$ is weakly increasing in a, so $A/q_1(a)$ is weakly decreasing in a while $q_1(a)+a$ is weakly increasing in a. The covariance between a weakly decreasing and a weakly increasing function of the same random variable is non-positive, and strictly negative whenever $q_1(a)$ is non-constant. Hence

$$q_0 \leq E[a+q_1(a)],$$

with strict inequality whenever $q_1(a)$ is non-constant.

Economic intuition. The date-0 price of capital is below its present discounted value even though the entrepreneur is unconstrained and risk-neutral at date 0. Capital is a particularly poor investment opportunity at date 0: when the date-1 productivity a is low, capital pays little in both flow dividend a and resale price $q_1(a)$. But a low a is precisely the state in which net worth and the resale price are low, so the marginal value of an extra unit of net worth at date 1, given by the shadow return $A/q_1(a)$ on a constrained entrepreneur, is highest. The payoff of date-0 capital is negatively correlated with the entrepreneur's investment opportunities at date 1, and the entrepreneur demands a risk premium even though her preferences are linear. She does not lever up against expected positive returns because the very states in which she would most need additional net worth are the states in which capital pays the least.